Merits:

- The $C.V$ is independent of the unit in which the measurement has been taken, but standard deviation depends on units of measurement. Hence one should use the coefficient of variation instead of the standard deviation.

Limitations:

- If the value of mean approaches 0, the coefficient of variation approaches infinity. So the minute changes in the mean will make major changes.

Example 6.12

If the coefficient of variation is 50 per cent and a standard deviation is 4, find the mean.

Solution:

$$\text{Coefficient of Variation} = \frac{\sigma}{\bar{x}} \times 100$$

$$50 = \frac{4}{\bar{x}} \times 100$$

$$\bar{x} = \frac{4}{50} \times 100 = 8$$

Example 6.13

The scores of two batsmen, A and B, in ten innings during a certain season, are as under:

A: Mean score = 50; Standard deviation = 5

B: Mean score = 75; Standard deviation = 25

Find which of the batsmen is more consistent in scoring.

Solution:

$$\text{Coefficient of Variation } (C.V) = \frac{\sigma}{\bar{x}} \times 100$$

$$C.V \text{ for batsman } A = \frac{5}{50} \times 100 = 10\%$$

$$C.V \text{ for batsman } B = \frac{25}{75} \times 100 = 33.33\%$$

The batsman with the smaller $C.V$ is more consistent.

Since for Cricketer A, the $C.V$ is smaller, he is more consistent than B.

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